SAURAV KISHOR AVACHAT

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www.linkedin.com/in/saurav-avachat-073778267 | saurav3103.github.io (Portfolio)

MSc Systems & Control student at TU Delft with experience in optimal control, state estimation, and signal processing, from Kalman-filter-based LQG design to PLC/DCS automation at Siemens and ABB and physics-guided ML for sensors. Interested in imaging systems and optics. Additional projects are on my portfolio website

EDUCATION

Delft University of Technology, Delft, Netherlands **August 2026 - August 2028**
Master of Science in Systems and Control
Delft Center for Systems and Control (DCSC)
Relevant Courses: Statistical Signal Processing, Control Theory, Modelling of Dynamical Systems, Optimization.

Visvesvaraya National Institute of Technology, Nagpur, India **November 2022 – May 2026**
Bachelor of Technology in Electronics and Communication Engineering **CGPA 9.18/10 | Rank: 3/120**
Minor Subject: VLSI ***(CGPA 9.53/10.0)***
Relevant Courses: Control Systems, Digital Signal Processing, Adaptive Signal Processing, Biomedical Signal Processing, Machine Learning.

INDUSTRY EXPERIENCE

Siemens Ltd., Navi Mumbai, India **January 2026 – May 2026**
Digitalization and Automation Intern | Siemens Digital Industries

- Engineered a real-time **S7-1200 Digital Twin** modelling discrete-state control logic for 9 industrial automation scenarios; enabled hardware-in-the-loop (HIL) simulation to validate state transitions before physical deployment, **reducing prototyping iteration cycles**.
- Built a browser-based simulation interface integrating **a PCS 7 DCS simulation layer**, cutting the need for physical test-bench access during control logic validation.
- Contributed to the **SiTrain LMS platform (multi-role architecture, Supabase backend, Android APK**), extending the digital twin's use beyond validation into hands-on operator training.

ABB Ltd., Bangalore, India **May 2025 – August 2025**
Distributed Control Systems and Automation Intern | Process Automation Division

- Developed and validated control logic in **ABB Control Builder** for a **cold rolling mill's distributed control system**, ensuring correctness against existing plant behavior before rollout.
- Migrated legacy operator interfaces to modern, **dynamic HMI displays** for the mill's process automation setup, integrating system diagnostics to **improve fault visibility and operator response time**.
- Collaborated with the Process Automation Division to align new HMI logic with **real-time DCS data flows** from the rolling mill process, ensuring seamless integration with existing plant infrastructure.

Godrej and Boyce Mfg. Ltd., Mumbai, India **May 2024 – July 2024**
Technical Embedded Systems Intern | Security Solutions Design Division

- Designed and prototyped control logic for a smart locking system on the **MSPM0G3507 (TI) microcontroller**, covering state management for **lock/unlock and access-control events**.
- Implemented **high-reliability serial communication protocols (UART/ I2C)** for secure data transfer between the lock module and host system, safeguarding against data corruption/tampering.

RESEARCH EXPERIENCE

Advanced Materials and Devices Laboratory, Jeonbuk National University, South Korea **October 2025 – June 2026**
Machine Learning Research Intern (Remote). Professor In-charge Dr Sadia Ameen and Dr. A.S Akhtar.

- Developed **physics-guided ML models (Random Forest, XGBoost, Physics-Guided XGBoost**) for electrochemical sensor response prediction, achieving **MAE = 0.0317, RMSE = 0.0572, and Pearson r = 0.9757** by embedding domain physical constraints directly into the learning pipeline.
- Designed an **ensemble modelling strategy combining classical ML (RF, XGBoost)** with physics-informed architectures, improving generalization over purely data-driven baselines for cyber-physical sensor systems.
- Collaborated remotely with Dr. Sadia Ameen and Dr. A.S. Akhtar's group to **translate raw electrochemical sensor data into a validated predictive pipeline**, contributing to a first co-authored manuscript **accepted in *Measurement* (Elsevier, Q1).**

MedIA Lab, Visvesvaraya National Institute of Technology, Nagpur, India **January 2025 – May 2026**
Student Undergraduate Researcher. Professor In-charge Dr Deep Gupta. | Undergraduate Thesis.

- Architected and benchmarked **7 Mask R-CNN encoder-decoder variants** for oropharyngeal tumor segmentation, achieving **Dice Score > 0.75 and Confidence Score > 0.95** across **T1–T4 tumor stages** in clinical imaging data.
- Designed a systematic encoder-decoder ablation study to isolate which architectural changes drove segmentation gains, enabling reproducible, **high-precision tumor boundary detection**.
- Delivered the complete **research pipeline - from model architecture through evaluation** - culminating in a full undergraduate thesis under Dr. Deep Gupta, advancing automated tools for clinical oncology imaging.

PUBLICATIONS

- Eun-Bi Kim, S. Avachat, S. Ameen, and A.S. Akhtar, "**Machine Learning-Assisted Optimization of High-Performance ZnO@Graphene Oxide Sensors for Trace Metal Ion Detection**" *Measurement*, 2026 (Elsevier, Q1) *accepted.*
- Eun-Bi Kim, S. Avachat, S. Ameen, and A.S. Akhtar, "**Physics-Guided Machine Learning Optimization of ZIF-8/Co–N–C Nanocomposite Sensors for Electrochemical Detection of Cr (III) and Hg (II)"** *Manuscript in Preparation*, 2026.

PROJECTS

Adaptive Optics-Inspired Control Simulation
Libraries: Python, NumPy, SciPy (Welch PSD, Kalman filter), Matplotlib.

- Built a closed-loop optical control simulation **modeling wavefront correction via a slit-lens-focal plane setup,** progressing through PI, static system identification (Pseudo-Inverse control), first-order actuator lag, and **LQG controllers to restore ROI illumination against a mechanical jerk disturbance**.
- Identified disturbance frequencies from **open-loop PSD (Welch's method)** and designed a **steady-state Kalman filter with a harmonic oscillator disturbance model** for feedforward cancellation, recovering mean ROI intensity **from 0.1224 (open loop) to 0.1501** under actuator lag where D+ control degraded (std 0.0004 → 0.0147).

Controls C++: State-Space Control and Estimation Library
Libraries: C++17, Eigen, CMake, Google Test.

- Implemented a from-scratch **C++/Eigen library for state-space control and estimation**, including **DARE-based LQR**, a **discrete-time Kalman filter**, and a **Luenberger observer via control-estimation duality** (reusing Ackermann's formula on the transposed system), structured as a multi-module library with header/implementation separation and CMake build configuration.
- Composed **LQR and Kalman filtering into a full LQG controller** following the **separation principle**, verified via an 8-file **Google Test suite** checking closed-loop stability (eigenvalue placement), estimator convergence from incorrect initial conditions, and true-state regulation using only noisy output measurements — no direct state access.

Control-Theoretic Equilibrium Selection (CTES): LQR as a Global Optimization Proxy
Libraries: MATLAB (Control System Toolbox).

- Developed a framework repurposing **LQR optimal control theory as a global optimization proxy** for multimodal nonlinear functions; at each local minimum, linearizes the landscape as gradient-flow dynamics and solves the **Algebraic Riccati Equation (ARE)** with a depth-encoded state cost — using **−trace(P) as a basin-depth energy proxy** to rank candidate equilibria and identify the global minimum
- Achieved a **Spearman ρ = 1.000** across 1D/2D Gaussian wells and Rastrigin functions (up to 11 wells) using a unified, sign-agnostic depth encoding; validated on 3-link planar robot IK selection (~60 candidate configurations), with PD-controlled convergence to the CTES-selected equilibrium in simulation.

IMU Orientation Estimation: Classical Filters vs Learned Sequence Models
Libraries: Python, PyTorch, NumPy, pandas, ahrs, bagpy. | Dataset Used: TUM VI ROS dataset

- Built an end-to-end pipeline that extracts **IMU and VRPN motion-capture data from a ROS bag**, normalizes units, and time-aligns the streams; implemented a **custom quaternion-state Extended Kalman Filter with accelerometer and magnetometer correction** and ran it alongside the Madgwick and Mahony filters sample-by-sample to estimate roll, pitch, and yaw.
- Trained **Transformer and LSTM sequence models to regress quaternions directly from 20-sample windows of accelerometer, gyroscope, and magnetometer data**, and **benchmarked all five estimators against motion-capture ground truth using per-axis RMSE in degrees**, correcting for the time offset introduced by windowed predictions.

SKILLS

- **Controls & Systems Theory:** LQR, LQG, Kalman Filtering (EKF, Complementary Filters), State Observers, Luenberger Observers, PID, Gain Scheduling, System Identification.
- **Software and Embedded:** C++17 (Eigen, CMake, GTest), Embedded C, Python (NumPy, SciPy), MATLAB/Simulink, Microcontrollers (TI MSPM0, Arduino, ESP32), Protocols (UART, I2C).
- **Machine Learning & Deep Learning**: Pytorch, TensorFlow, Scikit-learn, DL Models (VGG-16, VGG-19, UNETR, SwinUNETR, UNET, Mask R-CNN, Faster R-CNN).
- **Signal Processing and Estimation:** Adaptive Optical Signal Processing, Sensor Fusion(IMU), AHRS Libraries, DSP Toolchains (Python/MATLAB), Quaternion Algebra.
- **Industrial Automation:** TIA Portal, SINAMICS Startdrive, ABB Control Builder, WinCC, Industrial Protocols (RS-485, Fieldbus, Profinet).

REFERENCES

- Dr. Deep Gupta, Associate Professor, Dept. of Electronics and Communication Engineering, VNIT Nagpur Undergraduate thesis supervisor (MedIA Lab) | deepgupta@ece.vnit.ac.in
- Dr. Saugata Sinha, Assistant Professor, Dept. of Electronics and Communication Engineering, VNIT Nagpur Instructor, Signals and DSP, Machine Learning | saugata.sinha@ece.vnit.ac.in
- Dr. Sadia Ameen, Professor, Head of Advanced Materials and Devices Laboratory, Jeonbuk National University Research supervisor (remote ML research internship) | sadiaameen@jbnu.ac.kr